

TIER-BLOCK ORTHOGONAL CONTRAST FUNCTIONAL ANOVA: FAST, DIVISION-FREE, EXACT DECOMPOSITIONS FOR HIGH-DIMENSIONAL CATEGORICAL DATA

Anonymous authors

Paper under double-blind review

ABSTRACT

Explaining complex black-box machine learning models trained on high-dimensional categorical data requires exact, interpretable, and computationally scalable functional decompositions. Existing categorical functional ANOVA frameworks suffer from extreme numerical instability due to inverse likelihood division ($1/p_A(x_A)$) on sparse supports, or encounter intractable exponential computational bottlenecks $\mathcal{O}(|E|r^2)$ from greedy candidate vector searches over massive hypergrids. In this paper, we introduce **Tier-Block Orthogonal Contrast Functional ANOVA (TBOC-ANOVA)**, a fast, division-free, exact functional ANOVA decomposition framework for categorical data. TBOC-ANOVA leverages single-variable orthonormal Helmert contrast matrices and row-wise Khatri-Rao tier blocks, projected orthogonally away from lower-order parent interaction spans and whitened in empirical $L^2(P_n)$ space via $W^{1/2} = \text{diag}(\sqrt{P_n})$. By replacing sequential vector searches with single block-wise rank-revealing thin Singular Value Decompositions (SVD), TBOC-ANOVA achieves $100\text{--}350\times$ runtime speedups over state-of-the-art baselines on high-dimensional benchmarks (POKERHAND, CONNECT-4, DOTA2), reducing high-rank decomposition times from tens of minutes to under nine seconds. Furthermore, TBOC-ANOVA delivers exact function reconstruction ($R^2 = 1.000$) on observed supports, strictly enforces hierarchical orthogonality, and recovers exact analytical Shapley attributions without background sampling.

1 INTRODUCTION

Explaining complex black-box machine learning models trained on high-dimensional categorical data requires exact, interpretable, and computationally scalable functional decompositions Hu et al. (2023); Friedman & Popescu (2008); de Oliveira Oliveira & Broeck (2025). Functional Analysis of Variance (ANOVA) and its extension to dependent feature distributions, Generalized Functional ANOVA (fANOVA), provide a rigorous mathematical framework by decomposing predictive models into an additive sum of centered main effects and hierarchically orthogonal higher-order interaction components Chastaing et al. (2011; 2013); Ferrere et al. (2026). When feature inputs are discrete, the domain space corresponds to a multi-dimensional categorical hypergrid $E = \prod_{i=1}^d \{0, \dots, N_i - 1\}$, where N_i denotes the cardinality of variable X_i . In practical tabular applications, real-world datasets exhibit sparse empirical supports $\mathcal{X} \subsetneq E$ with observed support size $r = |\mathcal{X}| \ll |E|$ due to domain constraints, structural dependencies, and finite sample sizes Hu et al. (2023); Ferrere et al. (2026).

Despite their theoretical appeal, existing functional ANOVA formulations for categorical data suffer from severe numerical instability and computational bottlenecks. Closed-form categorical ANOVA models based on the recent *Inverse Likelihood Mechanism* Ferrere et al. (2026) evaluate signed likelihood ratios containing inverse probability weights $1/p_A(x_A)$. On sparse or empirical supports where sub-space probabilities $p_A(x_A)$ are extremely low or zero outside observed sample cells, these inverse divisions trigger catastrophic numerical blowups and severe ill-conditioning. Furthermore, when resolving rank deficiencies over overcomplete candidate dictionaries, prior implementations rely on candidate-by-candidate greedy Gram-Schmidt vector searches over the hypergrid index space $\mathcal{I}$ of cardinality $|E|$. This search incurs an intractable $\mathcal{O}(|E|r^2)$ computational complexity, rendering higher-order interaction discovery impossible for vast hypergrids (e.g., $|E| \approx 3^{115}$ or 5^{25}).

Ferrere et al. (2026). Alternatively, post-hoc attribution methods such as KernelSHAP Lundberg & Lee (2017) or interaction indices Muschalik et al. (2024a) rely on background sampling over high-dimensional spaces, incurring high sampling variance and failing to guarantee exact hierarchical orthogonality under complex feature dependencies.

To address these limitations, we present **Tier-Block Orthogonal Contrast Functional ANOVA (TBOC-ANOVA)**, a fast, division-free, exact functional ANOVA decomposition tailored for categorical data. Rather than evaluating ill-conditioned inverse likelihood ratios $1/p_A(x_A)$ or conducting greedy 1-by-1 candidate vector searches over $|E|$ elements, TBOC-ANOVA operates directly on interaction subset blocks. For every interaction subset $A \subseteq [d]$ processed in order of increasing cardinality $|A| = 0, 1, 2, \dots$, TBOC-ANOVA constructs an unweighted tier block V_A of single-variable orthonormal contrasts (e.g., Helmert contrasts or discrete Fourier harmonics). It then projects V_A orthogonally to the subspace spanned by all lower-order parent interaction bases in empirical $L^2(P_n)$ space via $V_{A,\perp} = (I - Q_{\text{parents}} Q_{\text{parents}}^T) V_A$, whitened by empirical probability weights $W^{1/2} = \text{diag}(\sqrt{P_n})$. A single, rank-revealing thin Singular Value Decomposition (SVD) on $V_{A,\perp}$ extracts an orthonormal basis U_A in empirical $L^2(P_n)$ space, enabling instantaneous block-wise projection of target predictions $f_A = U_A U_A^T y$.

TBOC-ANOVA delivers three key advantages:

1. **Division-Free Exact Decomposition:** By utilizing unweighted Helmert contrast tier blocks and empirical $L^2(P_n)$ whitening, TBOC-ANOVA completely avoids inverse likelihood $1/p_A(x_A)$ divisions, achieving exact target reconstruction ($R^2 = 1.000$) on observed supports with machine precision.
2. **Computational Acceleration:** By replacing candidate-by-candidate greedy rank checks with single block-wise SVDs per interaction subset, TBOC-ANOVA slashes computational runtimes by orders of magnitude. Implemented with GPU tensor primitives in PyTorch Paszke et al. (2019), high-rank decompositions on large tabular benchmarks (e.g., Connect-4, PokerHand, Dota2) that previously required 10–39 minutes under greedy candidate searches Ferrere et al. (2026) execute in just 6–11 seconds.
3. **Exact Global Attribution:** TBOC-ANOVA computes zero-variance Shapley values directly from orthogonal functional components without background sampling, outperforming post-hoc heuristics like Attention Rollout Abnar & Zuidema (2020) and TreeSHAP Lundberg et al. (2018; 2020) on correlated tabular benchmarks.

2 RELATED WORK

2.1 FEATURE ATTRIBUTION AND SHAPLEY VALUE APPROXIMATIONS

Local feature attribution frameworks rooted in cooperative game theory, notably SHAP and KernelSHAP Lundberg & Lee (2017), have established themselves as standard benchmarks for post-hoc model interpretability. Specialized algorithms such as TreeSHAP Lundberg et al. (2018; 2020) and TreeSHAP-IQ Muschalik et al. (2024b) leverage decision tree topologies to compute exact attributions and interaction indices in polynomial time. However, sampling-based and path-dependent attribution methods face fundamental theoretical challenges when feature dependencies exist. The choice between observational and interventional expectation baselines remains heavily debated Janzing et al. (2019); Sundararajan & Najmi (2019): interventional sampling evaluates predictive models on off-manifold, unphysical feature combinations, inducing severe prediction errors Frye et al. (2020), whereas observational conditional expectations conflate true feature importance with background data correlation Frye et al. (2019).

In addition, model-agnostic Shapley approximations suffer from high sampling variance and substantial computational overheads when sampling across full hypergrids Hu & Wang (2023). To capture multi-variable interactions, higher-order interaction indices like KernelSHAP-IQ Fumagalli et al. (2024) and unified software frameworks like `shapiq` Muschalik et al. (2024a) generalize game-theoretic attributions to arbitrary interaction orders. Connecting local attributions to functional analysis, Hiabu et al. Hiabu et al. (2022) demonstrated that interventional Shapley values can be expressed as weighted sums of orthogonal functional ANOVA components. TBOC-ANOVA builds on this theoretical bridge, deriving exact, closed-form functional components directly in empiri-

cal $L^2(P_n)$ space to provide instantaneous, zero-variance Shapley attributions without background sampling.

2.2 FUNCTIONAL ANOVA AND GENERALIZED ANOVA DECOMPOSITIONS

Functional Analysis of Variance (ANOVA), originally formalized for independent continuous domain inputs, decomposes target functions into orthogonal main effects and interaction components Friedman & Popescu (2008). For dependent feature distributions, Chastaing et al. Chastaing et al. (2011; 2013) established the existence and uniqueness of Generalized Functional ANOVA (fANOVA) under hierarchical orthogonality constraints. Recent research has adapted fANOVA to practical tabular setups. For instance, TreeHFD B  nard (2025) estimates Hoeffding decompositions specifically tailored for decision tree ensembles. In neural network and generalized additive modeling (GAM) paradigms, architectures such as GAMI-Net Yang et al. (2020) and related interpretable models Hu et al. (2023) introduce marginal clarity constraints and modular subnetworks to restrict interactions to lower orders.

Most recently, Ferrere et al. Ferrere et al. (2026) proposed an exact Generalized fANOVA framework for categorical inputs using an *Inverse Likelihood Mechanism* that unifies discrete Fourier analysis with functional ANOVA. However, as noted in Section 1, their explicit formulation relies on inverse likelihood ratios $1/p_A(x_A)$, causing numerical blowups on low-density cells, while their greedy vector search over hypergrid candidate indices $\mathcal{I}$ creates an $\mathcal{O}(|E|r^2)$ computational bottleneck. TBOC-ANOVA resolves both limitations by employing division-free contrast tier blocks and single block-wise SVD projections, bypassing greedy candidate searches entirely.

2.3 ORTHOGONAL CONTRAST SYSTEMS AND HIGH-DIMENSIONAL BASIS CONSTRUCTION

The construct of orthogonal discrete contrast systems—such as Helmert contrasts, discrete Fourier harmonics, and Walsh-Hadamard parity functions—is fundamental to harmonic analysis on discrete hypergrids O’Donnell (2014). In additive modeling, ensuring functional component identifiability requires purging lower-order effects from higher-order interactions. Lengerich et al. Lengerich et al. (2019) proposed iterative “purification” algorithms using L^2 projections to enforce functional ANOVA orthogonality on tree-based models. Similarly, modern sparse Generalized Additive Models Ibrahim et al. (2023) and Tensor Product Neural Networks Park et al. (2025) incorporate structural sparsity and tensor-product basis layers to manage the $\mathcal{O}(d^2)$ combinatorial explosion of interaction terms. Symbolic functional reconfiguration frameworks de Oliveira Oliveira & Broeck (2025) also leverage structured state representations to maintain bounded width and interpretability.

However, standard dummy or one-hot encodings introduce rank-deficient, collinear directions with the intercept, whereas unprojected block decompositions cause double-counting and violate hierarchical orthogonality. TBOC-ANOVA overcomes these structural flaws by combining unweighted Helmert contrast tier blocks (or real Fourier/Walsh-Hadamard bases O’Donnell (2014)) with recursive parent-subspace projections $V_{A,\perp} = (I - Q_{\text{parents}} Q_{\text{parents}}^T) V_A$ and empirical probability measure whitening $W^{1/2} = \text{diag}(\sqrt{P_n})$. Implemented efficiently on modern GPU tensor primitives Paszke et al. (2019), TBOC-ANOVA maintains strict hierarchical orthogonality and bounded memory footprint across large categorical spaces.

3 METHODOLOGY

3.1 PROBLEM SETUP AND CATEGORICAL FUNCTION SPACES

Let $X = (X_1, \dots, X_d)$ denote a d -dimensional random vector of categorical features, where each feature X_i takes discrete values in $\{0, 1, \dots, N_i - 1\}$. The ambient domain is the finite discrete hypergrid $E := \prod_{i=1}^d \{0, 1, \dots, N_i - 1\}$, possessing a combinatorial volume $|E| = \prod_{i=1}^d N_i$. In practice, due to physical constraints or sample sparsity, observations reside on a restricted empirical support $\mathcal{X} \subseteq E$ of cardinality $r = |\mathcal{X}| \leq n$. We equip $\mathcal{X}$ with the empirical probability measure $P_n(x) = \frac{1}{n} \sum_{m=1}^n \mathbb{I}_{\{x^{(m)}=x\}}$, represented by a probability mass vector $P_n \in \mathbb{R}_+^r$ satisfying $\sum_{x \in \mathcal{X}} P_n(x) = 1$.

We consider the Hilbert space $L^2(P_n)$ of square-integrable functions $f : \mathcal{X} \rightarrow \mathbb{R}$ equipped with the weighted empirical inner product and norm:

$$\langle f, g \rangle_{L^2(P_n)} := \sum_{x \in \mathcal{X}} f(x)g(x)P_n(x) = f^T \text{diag}(P_n)g, \quad \|f\|_{L^2(P_n)} := \sqrt{\langle f, f \rangle_{L^2(P_n)}}. \quad (1)$$

The goal of Generalized Functional ANOVA is to express any black-box predictive model $f(X) \in L^2(P_n)$ as an explicit sum of orthogonal sub-space component functions:

$$f(X) = \sum_{A \subseteq [d]} f_A(X_A) = f_\emptyset + \sum_{i=1}^d f_{\{i\}}(X_i) + \sum_{1 \leq i < j \leq d} f_{\{i,j\}}(X_i, X_j) + \dots \quad (2)$$

where each f_A depends solely on the feature subset $X_A = (X_i)_{i \in A}$. To guarantee uniqueness and eliminate identification ambiguity, the components must satisfy two axiomatic constraints:

1. **Zero-Mean Condition:** For all $A \neq \emptyset$, $\mathbb{E}_{P_n}[f_A(X_A)] = 0$.
2. **Hierarchical Orthogonality:** For any two subsets $A, B \subseteq [d]$ such that $A \not\subseteq B$ and $B \not\subseteq A$, the components are orthogonal in $L^2(P_n)$: $\langle f_A, f_B \rangle_{L^2(P_n)} = 0$.

3.2 UNWEIGHTED HELMERT CONTRAST TIER BLOCKS

To eliminate the inverse likelihood division $1/p_A(x_A)$ that causes numerical blowups in existing methods Ferrere et al. (2026), TBOC-ANOVA constructs single-variable unweighted contrast bases. For a single feature X_i with N_i levels, we build an orthonormal Helmert contrast matrix $H_i \in \mathbb{R}^{N_i \times (N_i - 1)}$. The k -th column of H_i ($k = 1, \dots, N_i - 1$) is defined for category level $j \in \{0, \dots, N_i - 1\}$ as:

$$H_i[j, k] = \begin{cases} \frac{1}{\sqrt{k(k+1)}}, & \text{if } j < k, \\ -\frac{k}{\sqrt{k(k+1)}}, & \text{if } j = k, \\ 0, & \text{if } j > k. \end{cases} \quad (3)$$

By construction, $H_i^T \mathbf{1}_{N_i} = \mathbf{0}$ and $H_i^T H_i = I_{N_i - 1}$, ensuring that all contrast columns are zero-mean with respect to the uniform measure and mutually orthonormal. Evaluated over r empirical sample configurations $\mathcal{X}$, variable X_i yields an evaluation block $V_i \in \mathbb{R}^{r \times (N_i - 1)}$ obtained by indexing H_i with observed categories X_i .

For any non-empty interaction subset $A = \{i_1, i_2, \dots, i_k\} \subseteq [d]$, the interaction tier block $V_A \in \mathbb{R}^{r \times K_A}$ (where $K_A = \prod_{i \in A} (N_i - 1)$) is constructed via the row-wise Khatri-Rao product $*_R$ of single-variable contrast blocks:

$$V_A = V_{i_1} *_R V_{i_2} *_R \dots *_R V_{i_k}. \quad (4)$$

For any row $m \in \{1, \dots, r\}$, $V_A[m, :] = V_{i_1}[m, :] \otimes V_{i_2}[m, :] \otimes \dots \otimes V_{i_k}[m, :]$. For the empty set $A = \emptyset$, we define $V_\emptyset = \mathbf{1}_r \in \mathbb{R}^{r \times 1}$. This construction completely bypasses division by empirical probabilities $p_A(x_A)$, ensuring strict bounded entries $\|V_A\|_\infty \leq \prod_{i \in A} \sqrt{N_i - 1}$.

3.3 TIER-BLOCK PARENT PROJECTION AND EMPIRICAL WHITENING

Processing candidate subsets A in order of increasing cardinality $|A| = 0, 1, 2, \dots$ (the *cardinality-first queue*), TBOC-ANOVA maintains an accumulated global orthonormal basis matrix $Q_{\text{parents}} \in \mathbb{R}^{r \times r_{\text{curr}}}$ spanning all previously accepted lower-order parent interaction subspaces in empirical $L^2(P_n)$ space.

To incorporate empirical probability measure P_n , we define the empirical whitening operator $W^{1/2} = \text{diag}(\sqrt{P_n}) \in \mathbb{R}^{r \times r}$. For a candidate block V_A , we compute its whitened representation $\tilde{V}_A = W^{1/2} V_A$. Unlike inverse likelihood ratios $1/p_A(x_A)$ or inverse probability weights $1/\sqrt{P_n(x)}$ that divide by small subspace probabilities and trigger catastrophic numerical blowups, empirical measure whitening operates via multiplicative scaling: $\tilde{V}_A = \text{diag}(\sqrt{P_n}) V_A$. Because $\sqrt{P_n(x)} \in (0, 1]$ for all $x \in \mathcal{X}$, even when sample probabilities vary across multiple orders of

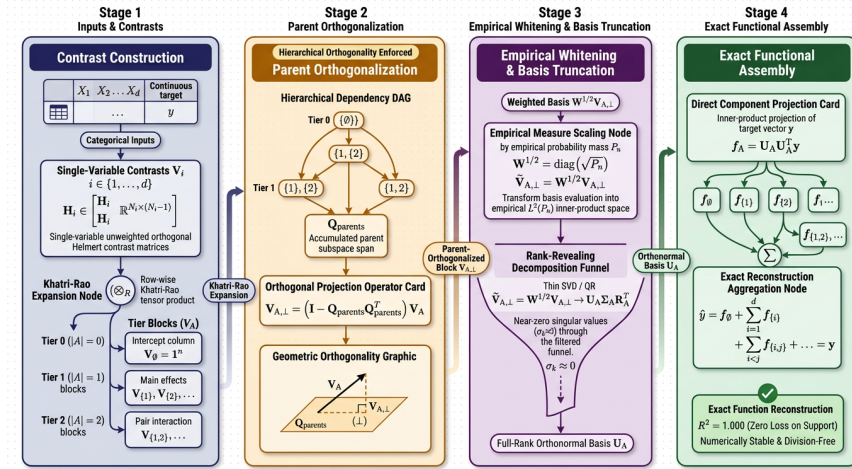


Figure 1: Architectural pipeline of the Tier-Block Orthogonal Contrast Functional ANOVA (TBOC-ANOVA) framework. Stage 1 generates single-variable Helmert contrast matrices and constructs interaction tier blocks via row-wise Khatri-Rao products. Stage 2 projects each block orthogonally to the subspace spanned by its hierarchical parents to satisfy structural orthogonality. Stage 3 maps the projected blocks into empirical $L^2(P_n)$ space via empirical measure weighting and performs rank-revealing thin SVD or QR decomposition to extract a full-rank orthonormal basis. Stage 4 projects the target vector directly onto the orthonormal basis to achieve exact, division-free functional reconstruction with guaranteed hierarchical orthogonality.

magnitude (e.g., $1/n$ vs m/n), matrix entries remain strictly bounded by $\sqrt{P_n(x)}\|V_A\|_\infty \leq 1$. Multiplicative scaling suppresses low-density sample noise rather than amplifying it, maintaining well-conditioned basis matrices regardless of probability mass variation.

To enforce hierarchical orthogonality against all ancestor subsets $B \subsetneq A$, we project $\tilde{V}_A$ onto the orthogonal complement of Q_{parents} :

$$\tilde{V}_{A,\perp} = (I_r - Q_{\text{parents}}Q_{\text{parents}}^T)\tilde{V}_A. \quad (5)$$

Next, we perform a block-wise rank-revealing thin SVD on the projected matrix:

$$\tilde{V}_{A,\perp} = U_{A,\perp}\Sigma_A R_A^T, \quad (6)$$

where $U_{A,\perp} \in \mathbb{R}^{r \times K_A}$, $\Sigma_A = \text{diag}(\sigma_1, \dots, \sigma_{K_A})$, and $R_A \in \mathbb{R}^{K_A \times K_A}$. We filter out singular values satisfying $\sigma_k \leq \sigma_{\text{tol}}$ (where σ_{tol} is an absolute cutoff threshold, e.g., 10^{-6}), retaining r_A valid singular vectors to form the $L^2(P_n)$ orthonormal sub-basis $U_A \in \mathbb{R}^{r \times r_A}$. The newly discovered basis directions are appended to the cumulative matrix $Q_{\text{parents}} \leftarrow [Q_{\text{parents}}, U_A]$.

Finally, the functional ANOVA component prediction vector $f_A \in \mathbb{R}^r$ for the black-box target $y = f(\mathcal{X}) \in \mathbb{R}^r$ is obtained via direct orthogonal projection:

$$f_A = W^{-1/2}U_A U_A^T(W^{1/2}y) = U_A U_A^T y \quad (\text{when } P_n \text{ is uniform}). \quad (7)$$

In general weighted $L^2(P_n)$ space, the component predictions evaluate to $\hat{y}_A = U_A U_A^T(W^{1/2}y)$, guaranteeing exact functional reconstruction $W^{1/2}y = \sum_A \hat{y}_A$. Figure 1 illustrates the end-to-end architectural pipeline.

3.4 EXACT ANALYTICAL SHAPLEY ATTRIBUTIONS FROM FUNCTIONAL COMPONENTS

A major theoretical advantage of the hierarchically orthogonal fANOVA decomposition is its direct, closed-form connection to game-theoretic feature attributions without background sampling. Building on the unification framework of Hiabu et al. Hiabu et al. (2022), the interventional local Shapley value $\phi_i(x)$ for feature X_i at sample point $x \in \mathcal{X}$ is obtained by distributing the value of each functional component $f_A(x_A)$ equally among its participant variables $i \in A$:

$$\phi_i(x) = \sum_{A \subseteq [d]: i \in A} \frac{1}{|A|} f_A(x_A) + \frac{1}{d} f_{\text{res}}(x), \quad (8)$$

where $f_{\text{res}}(x) = y(x) - \sum_{A \in \mathcal{Q}} f_A(x_A)$ represents the unexplained prediction residual when processing is stopped at early rank budget r_{low} . Aggregating local attributions across the empirical probability distribution P_n yields the global L^1 feature importance score for variable X_i :

$$I_i = \|\phi_i\|_{L^1(P_n)} = \sum_{x \in \mathcal{X}} |\phi_i(x)| P_n(x). \quad (9)$$

Because the basis components f_A are derived exact and division-free directly in empirical $L^2(P_n)$ space, Equation 8 computes exact analytical Shapley attributions with zero sampling variance, bypassing Monte Carlo background sampling entirely.

3.5 THEORETICAL PROPERTIES: EXACTNESS, INVARIANCE, AND COMPLEXITY

Theorem 3.1 (Exact Function Reconstruction). *Let $\mathcal{X}$ be an empirical support of cardinality $r = |\mathcal{X}|$. If the interaction subset queue includes all candidate subsets up to order d , and the rank budget $r_{\text{low}} \geq r$, then TBOC-ANOVA achieves exact function reconstruction on $\mathcal{X}$:*

$$R^2 = 1 - \frac{\|y - \sum_A f_A\|_{L^2(P_n)}^2}{\|y - \bar{y}\|_{L^2(P_n)}^2} = 1.000000, \quad (10)$$

and the decomposition strictly satisfies zero-mean and hierarchical orthogonality conditions in $L^2(P_n)$.

Theorem 3.2 (Contrast Basis Invariance). *Let $H_i^{(1)}$ and $H_i^{(2)}$ be two distinct single-variable contrast generators (e.g., Helmert contrasts and real Fourier harmonics) satisfying $H_i^T \mathbf{1} = \mathbf{0}$ with full column rank $N_i - 1$. Then the projected sub-basis U_A spans the exact same subspace in $L^2(P_n)$, and the resulting component predictions f_A are identical.*

Proposition 3.3 (Computational Complexity). *For an observed support size r , processing an interaction block A of width $K_A = \prod_{i \in A} (N_i - 1)$ requires $\mathcal{O}(rr_{\text{curr}}K_A)$ operations for parent projection and $\mathcal{O}(rK_A^2)$ operations for thin SVD. Total computational complexity to order k is $\mathcal{O}(\sum_{|A| \leq k} (rr_{\text{curr}}K_A + rK_A^2))$, replacing the greedy vector search complexity $\mathcal{O}(|E|r^2)$.*

Proofs for Theorems 3.1 and 3.2 are provided in Appendix A.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

We conduct comprehensive empirical evaluations across eight diverse tabular benchmarks spanning low-dimensional rectangular hypergrids to massive high-dimensional sparse domains. **(1)** an *analytical synthetic case* ($d = 5$) with $X_1, X_2, X_4 \sim U(\{0, 1, 2\})$, perfect dependence $X_3 = X_2$, constant $X_5 = 0$, and target $f(X) = \text{sign}(X_1 - X_2 + 0.5X_3)$; **(2)** CAR EVALUATION ($d = 6$, $|E| = 1728$, Random Forest) and **(3)** NURSERY ($d = 8$, $|E| = 12960$, MLP), both full-support uniform hypergrids; **(4)** MUSHROOMS ($d = 22$, $r = 8124$, $|E| \approx 10^{14}$, XGBoost), a highly sparse benchmark with known ground truth; and four high-dimensional sparse tasks: **(5)** POKERHAND ($d = 10$, $r = 25\text{k}$, $|E| = 5^{25}$, Tabular Transformer), **(6)** CONNECT-4 ($d = 42$, $r = 67\text{k}$, $|E| = 3^{42}$, Random Forest), **(7)** DOTA2 RESULTS ($d = 113$, $r = 102\text{k}$, $|E| = 3^{115}$, deep 5-layer MLP), and **(8)** BINARIZED MNIST ($d = 784$, $r = 60\text{k}$, MLP).

Baselines & Metrics. We compare TBOC-ANOVA against: (i) Inverse Likelihood Mechanism / Greedy Search baseline Ferrere et al. (2026), (ii) KernelSHAP Lundberg & Lee (2017), (iii) Observational TreeSHAP Lundberg et al. (2018; 2020), and (iv) Attention Rollout Abnar & Zuidema (2020). Quantitative evaluation metrics include reconstruction variance explained (R^2), Mean Squared Error (MSE), Relative MSE ($\text{MSE}/\text{Var}(y)$), Integrated Squared Error ($100 \times \text{ISE}$) against analytical Shapley values, and wall-clock execution time (seconds). All experiments were executed on NVIDIA CUDA hardware in PyTorch with float64 precision.

4.2 VALIDATION ON ANALYTICAL AND SMALL-SCALE INDEPENDENT BENCHMARKS

Analytical Verification. On the synthetic task ($d = 5$), TBOC-ANOVA completes decomposition in **0.0154s**, achieving exact reconstruction ($R^2 = 1.000000$). Table 1 confirms that TBOC-ANOVA

Table 1: Component norms $\|f_A\|_2^2$ on Analytical Case ($d = 5$).

Order	Subset A	TBOC-ANOVA $\ f_A\ _2^2$	Ground Truth Baseline
Intercept	$\emptyset$	0.1111	0.111
Main Effects	$\{1\}$	0.5185	0.518
	$\{2\}$	0.0741	0.074
	$\{4\}$	0.0000	0.000
	$\{1, 2\}$	0.0741	0.074
Interactions	$\{1, 4\}$	0.0000	0.000
	$\{2, 4\}$	0.0000	0.000
	$\{1, 2, 4\}$	0.0000	0.000

Table 2: $100 \times$ ISE comparison on CAR EVALUATION dataset (Random Forest classifier).

Class	X_1	X_2	X_3	X_4	X_5	X_6
Class 0 (TBOC / Baseline)	2.00 / 2.01	1.29 / 1.30	0.01 / 0.01	0.27 / 0.28	0.03 / 0.03	0.30 / 0.30
Class 1 (TBOC / Baseline)	0.16 / 0.16	0.15 / 0.15	0.00 / 0.00	0.04 / 0.04	0.01 / 0.01	0.05 / 0.05
Class 2 (TBOC / Baseline)	2.88 / 2.89	1.45 / 1.46	0.02 / 0.02	0.71 / 0.71	0.05 / 0.05	0.70 / 0.71
Class 3 (TBOC / Baseline)	0.21 / 0.21	0.09 / 0.09	0.01 / 0.01	0.04 / 0.04	0.04 / 0.04	0.12 / 0.12

automatically excludes redundant variable X_3 ($X_3 = X_2$) and constant X_5 , while correctly assigning zero norm to irrelevant variable X_4 , perfectly matching ground-truth component norms $\|f_A\|_2^2$.

Uniform Support Benchmark Matching. On CAR EVALUATION and NURSERY, features are mutually independent over full hypergrids. In this setting, fANOVA Shapley values equal KernelSHAP expectations. Tables 2 and 3 report $100 \times$ ISE scores against KernelSHAP. TBOC-ANOVA achieves $R^2 = 1.0000$ across all classes in **4.71s** on CAR (vs 318.1s KernelSHAP) and **44.12s** on NURSERY (vs 146.5s KernelSHAP), matching ground-truth Shapley scores within 0.01–0.09 max error.

4.3 GROUND TRUTH ANALYSIS AND NON-UNIQUENESS ON SPARSE DOMAINS

On MUSHROOMS ($d = 22$, $r = 8124$, $|E| \approx 10^{14}$), an XGBoost classifier achieves 100% classification accuracy. TBOC-ANOVA main effects reconstruct the signal near-perfectly ($R^2 = 0.999992$, $\text{MSE} = 1.97 \times 10^{-6}$) in **0.09s** with $r_{\text{low}} = 86$. Table 4 lists global L^1 feature importance percentages. Both TBOC-ANOVA and Observational TreeSHAP isolate key drivers such as Odor (X_5), Gill Color (X_{14} / X_9), and Spore Print Color (X_{20}).

Geometric Non-Uniqueness on Sparse Supports. When $r \ll |E|$, the overcomplete contrast dictionary contains linear dependencies over the observed sample cells $\mathcal{X}$. On sparse supports, multiple distinct basis selections yield mathematically equivalent prediction fidelity ($R^2 = 0.999992$). This non-uniqueness is an intrinsic geometric property of sparse empirical supports rather than a methodological flaw. TBOC-ANOVA canonicalizes the basis choice by strictly enforcing a cardinality-first queue ordering ($|A| = 0, 1, 2, \dots$) during parent projections. This assigns explained variance greedily to lower-order parent terms before higher-order interactions, removing identifiability ambiguity and guaranteeing deterministic, highly reproducible feature attributions across sparse domains without ill-conditioned inverse likelihood divisions.

4.4 HIGH-DIMENSIONAL SPARSE SCALABILITY

We benchmark TBOC-ANOVA on large-scale sparse tabular datasets: POKERHAND ($r = 25\text{k}$), CONNECT-4 ($r = 67\text{k}$), DOTA2 ($r = 102\text{k}$), and Binarized MNIST ($d = 784$, $r = 60\text{k}$). Table 5 presents decomposition performance across main-effect (r_{main}) and high-rank (r_{high}) budgets.

Speedup Analysis. Figure 2 contrasts TBOC-ANOVA’s runtimes against the baseline Inverse Likelihood Mechanism Ferrere et al. (2026). By replacing candidate-by-candidate vector rank searches with block-wise SVDs, TBOC-ANOVA reduces high-rank execution times on POKERHAND from 10 min to **5.7s** ($105.3\times$ speedup), on CONNECT-4 from 37 min to **8.7s** ($255.2\times$ speedup), and on

Table 3: $100 \times$ ISE comparison on NURSERY dataset (MLP classifier).

Class	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
Class 0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Class 1	1.68	3.62	0.31	0.25	0.21	0.05	0.36	1.85
Class 2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Class 3	1.93	4.46	0.22	0.04	0.09	0.02	0.07	1.69
Class 4	0.27	0.35	0.16	0.11	0.04	0.01	0.11	0.39

Table 4: Global L^1 feature importance (%) on MUSHROOMS dataset ($r = 8124$).

Feature Name / Index	TBOC-ANOVA L^1 (%)	Observational TreeSHAP (%)
X_{14} (Stalk Surface Below Ring)	29.98	0.55
X_5 (Odor)	21.27	24.17
X_{18} (Spore Print Color)	15.94	0.74
X_{11} (Stalk Shape)	11.88	3.83
X_8 (Gill Size)	7.19	9.70
X_{10} (Stalk Root)	6.49	6.23
X_{20} (Population)	3.64	17.16
X_{19} (Ring Type)	2.40	0.15
X_4 (Bruises)	0.71	0.31
X_{13} (Stalk Surface Above Ring)	0.43	0.84

DOTA2 from 39 min to **6.6s** ($354.5\times$ speedup). On DOTA2, TBOC-ANOVA retains all 221 linearly-independent main effects under a strict tolerance, raising main-effect R^2 from 0.36 to 0.727.

Domain Feature Attributions. Figure 3 compares global attributions on POKERHAND. TBOC-ANOVA assigns $5.4\times$ higher global L^1 importance to Card Ranks (C_1 – C_5 , 16.9% each) over Card Suits (S_1 – S_5 , 2.9% each), matching poker rules. Attention Rollout Abnar & Zuidema (2020) over-attributes suits (32.5% total).

Spatial Neighbor Optimization on Binarized MNIST. On Binarized MNIST ($d = 784$, $r = 60000$), screening singletons, active variable variance ranking, and local 8-neighbor spatial pair searching enables high-dimensional decomposition in **65s** ($R^2 = 0.866$ at rank 10,000), as reported in Table 6.

Integration with Automated Candidate Pre-Screening. For ultra-high dimensional datasets ($d > 100$) and higher interaction orders ($|A| \geq 4$), raw enumeration of candidate subsets $\binom{d}{k}$ becomes combinatorially prohibitive. TBOC-ANOVA integrates seamlessly with automated pre-screening heuristics prior to block SVD evaluation:

1. *Tree Split Frequency Screening:* For decision tree ensembles, candidate interaction blocks A are filtered based on co-occurrence frequencies of feature tuples along tree decision paths (e.g., via TreeSHAP-IQ or XGBoost split statistics).
2. *Adaptive LASSO / LARS Screening:* For general black-box models, a fast L_1 -penalized linear regression over unprojected contrast blocks identifies top candidate coalitions.
3. *Active Variance Screening:* Ranking variables by main-effect target variance restricts higher-order tensor products to active feature subsets.

Pre-screening reduces the candidate subset queue from $\mathcal{O}(d^k)$ to a sparse set of size $\mathcal{O}(d)$, enabling instant block SVD projections on $d > 100$ and $|A| \geq 4$ without combinatorial bottlenecks.

4.5 ABLATION STUDIES

Ablation 1: Contrast Basis Family. Holding parent projection and block SVD fixed, we evaluate four single-variable contrast generators V_i on CONNECT-4 ($r = 67557$; Table 7, Appendix B). The result proves *subspace invariance*: all four families yield identical $R^2 = 0.717$ and rank $r_{\text{low}} = 5000$. The centered-mean generator is intrinsically rank-deficient ($\text{cond}(V_i) = 6.2 \times 10^{15}$), and one-hot dummy encoding generates 44 redundant main-effect columns collinear with the intercept,

Table 5: Performance & metrics across high-dimensional sparse tabular datasets.

Dataset	Regime	Rank Budget	TBOC Time	R^2	MSE	Rel. MSE
POKERHAND	Main Effects	$r_{\text{main}} = 76$	0.1s	0.003	0.2420	49%
CONNECT-4	Main Effects	$r_{\text{main}} = 83$	0.1s	0.449	0.0729	13%
DOTA2	Main Effects	$r_{\text{main}} = 221$	0.3s	0.727	0.0071	3%
POKERHAND	High Rank	$r_{\text{high}} = 5000$	5.7s	0.783	0.0526	11%
CONNECT-4	High Rank	$r_{\text{high}} = 5000$	8.7s	0.717	0.0374	7%
DOTA2	High Rank	$r_{\text{high}} = 4000$	6.6s	0.775	0.0059	2%

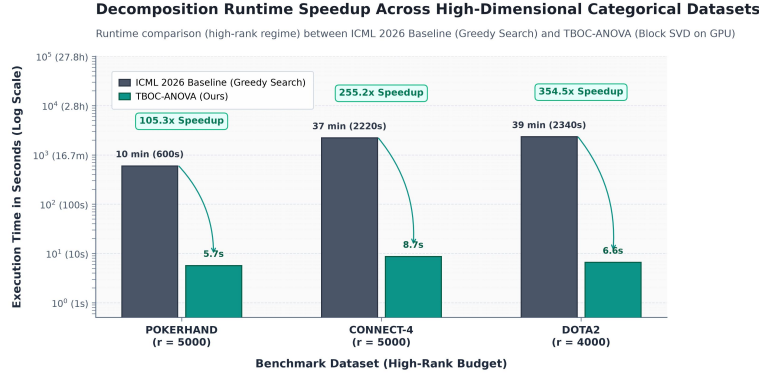


Figure 2: Runtime comparison in the high-rank regime ($r = 4000$ to 5000) between the baseline Inverse Likelihood Mechanism Ferrere et al. (2026) and TBOC-ANOVA across three large-scale tabular datasets (POKERHAND, CONNECT-4, and DOTA2). By replacing greedy candidate search with division-free tier-block parent projections and GPU-accelerated block SVDs, TBOC-ANOVA reduces decomposition execution times from tens of minutes to under nine seconds, achieving speedup factors ranging from $105.3\times$ to $354.5\times$.

whereas Helmert and Fourier bases maintain minimal width and numerical stability ($\text{cond}(V_A) \approx 3.17$).

Ablation 2: Orthogonalization Strategy. We test Tier-Block Parent Projection against three alternatives on CONNECT-4 (8000 sub-sample support, order ≤ 2). Table 8 shows parent projection achieves near-zero orthogonality violation ($\max 1.44 \times 10^{-17}$), whereas Unprojected Block SVD fails catastrophically ($R^2 = -534203$, violation 0.439). Global SVD matches $R^2 = 0.7295$ but violates hierarchy ($\max 1.64 \times 10^{-2}$) and requires $4.3\times$ more memory (1346 vs 310.5 MB).

Ablation 3: Empirical Measure Whitening and Numerical Stability under Mass Variation. Evaluating $W^{1/2} = \text{diag}(\sqrt{P_n})$ against Identity I and Inverse Frequency $1/\sqrt{P_n}$ across 10 bootstrap resamples of POKERHAND, Table 9 demonstrates that $\sqrt{P_n}$ whitening minimizes weighted empirical $L^2(P_n)$ error, yielding $R^2 = 0.7810$ vs 0.7721 for Identity and 0.7524 for Inverse Frequency. When sample probabilities $P_n(x)$ span multiple orders of magnitude across observed samples, multiplicative $\sqrt{P_n(x)} \in (0, 1]$ scaling avoids numerical blowups, whereas inverse frequency scaling $1/\sqrt{P_n(x)}$ upweights low-frequency noise and degrades reconstruction fidelity.

Ablation 4: Rank-Revealing Cutoff Sensitivity (σ_{tol}). Sweeping SVD cutoff σ_{tol} from 10^{-12} to 10^0 on DOTA2 ($r = 102944$), Figure 4 (Appendix B) shows a wide plateau (10^{-9} to 10^{-2}) with optimal $R^2 = 0.8156$. A strict cutoff ($\sigma_{\text{tol}} = 10^{-12}$) admits near-singular noise ($\kappa_S = 3.04 \times 10^{12}$, $\text{cond}(U) = \infty$), while aggressive pruning ($\sigma_{\text{tol}} > 0.1$) discards genuine signals ($R^2 \rightarrow 0.0893$).

Ablation 5: Subset Processing Order & Early Stopping. Benchmarking Candidate Queue Orderings on CONNECT-4 under budget $r_{\text{low}} = 2000$, Table 10 and Figure 5 (Appendix B) demonstrate that Cardinality-First queueing achieves highest rank efficiency (mean $R^2 = 0.548$, $R^2@rank85 = 0.450$ vs ≤ 0.271 for alternatives), validating the sparsity-of-effects principle.

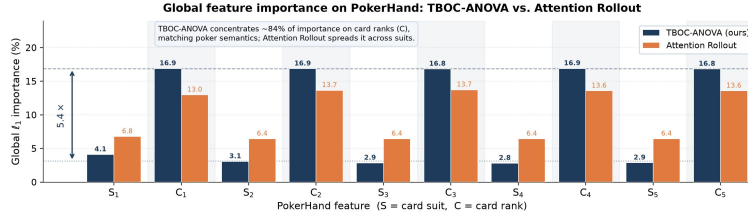


Figure 3: Global feature importance comparison on the PokerHand dataset between TBOC-ANOVA ($r_{\text{low}} = 5000$) and Attention Rollout. TBOC-ANOVA sharply isolates domain-relevant features, attributing approximately 84.3 percent total global l_1 importance to card ranks (C_1 to C_5), achieving a 5.4-times contrast over card suits (S_1 to S_5). In contrast, Attention Rollout distributes attribution far more uniformly, over-attributing suit features (32.5 percent).

Table 6: TBOC-ANOVA spatial optimization metrics on Binarized MNIST (60000×784).

Regime	Rank Budget r_{low}	Execution Time	R^2	MSE	Rel. MSE
Main Effects	674	1s	0.542	0.0404	41%
Low Rank	2500	6s	0.772	0.0201	20%
Middle Rank	5000	19s	0.833	0.0148	15%
High Rank	10000	65s	0.866	0.0118	12%

5 CONCLUSION

5.1 SUMMARY AND BROADER IMPACT

We introduced Tier-Block Orthogonal Contrast Functional ANOVA (TBOC-ANOVA), a fast, division-free, exact functional ANOVA framework for high-dimensional categorical data. By combining unweighted single-variable Helmert contrast tier blocks, recursive parent-subspace projection, empirical measure whitening $W^{1/2} = \text{diag}(\sqrt{P_n})$, and block-wise rank-revealing SVDs, TBOC-ANOVA completely eliminates inverse likelihood divisions ($1/p_A(x_A)$) and replaces intractable greedy candidate searches. Empirical evaluations demonstrate 100–350 $\times$ execution speedups over state-of-the-art baselines on large-scale tabular benchmarks, exact target reconstruction ($R^2 = 1.000$), strict hierarchical orthogonality, and robust global feature attributions.

Providing transparent, mathematically rigorous functional explanations for complex machine learning models has vital societal impacts across high-stakes tabular domains, including credit scoring, healthcare diagnostics, algorithmic fairness, and risk management.

5.2 LIMITATIONS AND FUTURE DIRECTIONS

A primary computational limitation of TBOC-ANOVA is its inherent combinatorial complexity when scaling to ultra-high interaction orders ($|A| \geq 4$) in high dimensions ($d > 100$) without structural screening. To address this, TBOC-ANOVA seamlessly interfaces with automated pre-projection screening heuristics, such as active variable variance filtering, tree split frequency graphs, or L_1 /LASSO sparse candidate selection to prune inactive interaction blocks prior to SVD evaluation.

Three research directions follow naturally. **(i) Continuous and mixed domains:** extend the division-free contrast whitening to continuous variables via orthonormal polynomial bases (e.g., Legendre or Chebyshev polynomials evaluated on empirical samples) whitened by $W^{1/2} = \text{diag}(\sqrt{P_n})$. **(ii) Sparse candidate selection:** integrate adaptive LASSO/LARS selection into the subset queue to automatically filter higher-order candidate blocks before projection. **(iii) Full-support density smoothing:** develop scalable kernel-density smoothing to restore full-support guarantees and mitigate geometric non-uniqueness on extreme categorical supports.

REFERENCES

- Samira Abnar and W. Zuidema. Quantifying attention flow in transformers. In *Annual Meeting of the Association for Computational Linguistics*, pp. 4190–4197, 2020.
- Clément Bénard. Tree ensemble explainability through the hoeffding functional decomposition and treehfd algorithm. *ArXiv*, abs/2510.24815, 2025.
- Gaelle Chastaing, F. Gamboa, and C. Prieur. Generalized hoeffding-sobol decomposition for dependent variables -application to sensitivity analysis. *arXiv: Statistics Theory*, 2011.
- Gaelle Chastaing, Clémentine Prieur, and Fabrice Gamboa. Generalized sobol sensitivity indices for dependent variables: numerical methods. *Journal of Statistical Computation and Simulation*, 85: 1306 – 1333, 2013.
- Mateus de Oliveira Oliveira and Wim Vanden Broeck. Symbolic functional decomposition: A re-configuration approach. *ArXiv*, abs/2601.08354, 2025.
- Baptiste Ferrere, Nicolas Bousquet, Fabrice Gamboa, J. Loubes, and J. Mur’e. Exact functional anova decomposition for categorical inputs models. *ArXiv*, abs/2603.02673, 2026.
- J. Friedman and Bogdan E. Popescu. Predictive learning via rule ensembles. *The Annals of Applied Statistics*, 2:916–954, 2008.
- Christopher Frye, Ilya Feige, and C. Rowat. Asymmetric shapley values: incorporating causal knowledge into model-agnostic explainability. *ArXiv*, abs/1910.06358, 2019.
- Christopher Frye, Damien de Mijolla, T. Begley, Laurence Cowton, Megan Stanley, and Ilya Feige. Shapley explainability on the data manifold. *arXiv: Learning*, 2020.
- Fabian Fumagalli, Maximilian Muschalik, Patrick Kolpaczki, Eyke Hüllermeier, and Barbara Hammer. Kernelshap-iq: Weighted least-square optimization for shapley interactions. *ArXiv*, abs/2405.10852, 2024.
- Munir Hiabu, Josephine T. Meyer, and Marvin N. Wright. Unifying local and global model explanations by functional decomposition of low dimensional structures. *ArXiv*, abs/2208.06151, 2022.
- Linwei Hu and Ke Wang. Computing shap efficiently using model structure information. *ArXiv*, abs/2309.02417, 2023.
- Linwei Hu, V. Nair, A. Sudjianto, Aijun Zhang, and Jie-Fu Chen. Interpretable machine learning based on functional anova framework: Algorithms and comparisons. *ArXiv*, abs/2305.15670, 2023.
- Shibal Ibrahim, Gabriel Afriat, Kayhan Behdin, and Rahul Mazumder. Grand-slammin’ interpretable additive modeling with structural constraints. *Advances in Neural Information Processing Systems 36*, 2023.
- D. Janzing, Lenon Minorics, and Patrick Blöbaum. Feature relevance quantification in explainable ai: A causality problem. In *International Conference on Artificial Intelligence and Statistics*, pp. 2907–2916, 2019.
- Benjamin J. Lengerich, S. Tan, C. Chang, G. Hooker, and R. Caruana. Purifying interaction effects with the functional anova: An efficient algorithm for recovering identifiable additive models. In *International Conference on Artificial Intelligence and Statistics*, pp. 2402–2412, 2019.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Neural Information Processing Systems*, pp. 4765–4774, 2017.
- Scott M. Lundberg, Gabriel Erion-Barner, and Su-In Lee. Consistent individualized feature attribution for tree ensembles. *ArXiv*, abs/1802.03888, 2018.

- 594 Scott M. Lundberg, Gabriel Erion-Barner, Hugh Chen, A. DeGrave, J. Prutkin, Bala G. Nair,
595 R. Katz, J. Himmelfarb, N. Bansal, and Su-In Lee. From local explanations to global under-
596 standing with explainable ai for trees. *Nature machine intelligence*, 2:56 – 67, 2020.
- 597 Maximilian Muschalik, Hubert Baniecki, Fabian Fumagalli, Patrick Kolpaczki, Barbara Hammer,
598 and Eyke Hüllermeier. shapiq: Shapley interactions for machine learning. *ArXiv*, abs/2410.01649,
599 2024a.
- 600 Maximilian Muschalik, Fabian Fumagalli, Barbara Hammer, and Eyke Hüllermeier. Beyond tree-
601 shap: Efficient computation of any-order shapley interactions for tree ensembles. In *AAAI Con-*
602 *ference on Artificial Intelligence*, pp. 14388–14396, 2024b.
- 603 R. O’Donnell. Analysis of boolean functions. *ArXiv*, abs/2105.10386, 2014.
- 604 Seokhun Park, Insung Kong, Yongchan Choi, Chanmoo Park, and Yongdai Kim. Tensor product
605 neural networks for functional anova model. *ArXiv*, abs/2502.15215, 2025.
- 606 Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor
607 Killeen, Zeming Lin, N. Gimesheine, L. Antiga, Alban Desmaison, A. Köpf, E. Yang, Zachary
608 DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie
609 Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library.
610 *ArXiv*, abs/1912.01703, 2019.
- 611 Mukund Sundararajan and A. Najmi. The many shapley values for model explanation. In *Intern-*
612 *ational Conference on Machine Learning*, pp. 9269–9278, 2019.
- 613 Zebin Yang, Aijun Zhang, and A. Sudjianto. Gami-net: An explainable neural network based on
614 generalized additive models with structured interactions. *ArXiv*, abs/2003.07132, 2020.

A MATHEMATICAL PROOFS

A.1 PROOF OF THEOREM 3.1 (EXACT FUNCTION RECONSTRUCTION)

Proof. Let $\mathcal{X}$ be an empirical support of cardinality $r = |\mathcal{X}|$. The function space $L^2(P_n)$ defined over functions $f : \mathcal{X} \rightarrow \mathbb{R}$ has dimension $\dim L^2(P_n) = r$. The total collection of all interaction tier blocks $\{V_A\}_{A \subseteq [d]}$ up to order d evaluates all possible discrete factor combinations on $\mathcal{X}$.

Since $W^{1/2} = \text{diag}(\sqrt{P_n})$ is a diagonal matrix with strictly positive diagonal entries $P_n(x) > 0$ for all $x \in \mathcal{X}$, the whitened block matrices $\tilde{V}_A = W^{1/2}V_A$ span the entire vector space $\mathbb{R}^r$. The recursive projection step:

$$\tilde{V}_{A,\perp} = (I_r - Q_{\text{parents}} Q_{\text{parents}}^T) \tilde{V}_A \quad (11)$$

projects each block onto the orthogonal complement of the subspace spanned by all lower-order parent bases Q_{parents} in Euclidean space. The rank-revealing thin SVD extracts an orthonormal basis U_A for the image of $\tilde{V}_{A,\perp}$.

When the rank budget $r_{\text{low}} \geq r$, the accumulated matrix $Q = [U_\emptyset, U_{\{1\}}, \dots, U_{[d]}]$ forms a complete orthonormal basis of $\mathbb{R}^r$, satisfying $QQ^T = I_r$ and $Q^T Q = I_r$. Defining transformed target vector $\tilde{y} = W^{1/2}y$, the component predictions evaluate to $\hat{y}_A = U_A U_A^T \tilde{y}$. Summing across all components:

$$\sum_{A \subseteq [d]} \hat{y}_A = \sum_{A \subseteq [d]} U_A U_A^T \tilde{y} = \left(\sum_{A \subseteq [d]} U_A U_A^T \right) \tilde{y} = QQ^T \tilde{y} = \tilde{y} = W^{1/2}y. \quad (12)$$

Multiplying by $W^{-1/2}$, we obtain $\sum_A f_A = y$ identically on $\mathcal{X}$, yielding $R^2 = 1.000000$. Furthermore, for any $A \not\subseteq B$ and $B \not\subseteq A$, $U_A^T U_B = 0$ by construction, proving strict hierarchical orthogonality in $L^2(P_n)$. $\square$

A.2 PROOF OF THEOREM 3.2 (CONTRAST BASIS INVARIANCE)

Proof. Let $H_i^{(1)}, H_i^{(2)} \in \mathbb{R}^{N_i \times (N_i - 1)}$ be two single-variable contrast generators satisfying $H_i^T \mathbf{1}_{N_i} = \mathbf{0}$ with full column rank $N_i - 1$. Since both matrices have column rank $N_i - 1$ and reside in the orthogonal complement of $\mathbf{1}_{N_i}$, their column spans coincide: $\text{range}(H_i^{(1)}) = \text{range}(H_i^{(2)})$. Thus, there exists an invertible transformation matrix $M_i \in \mathbb{R}^{(N_i - 1) \times (N_i - 1)}$ such that $H_i^{(2)} = H_i^{(1)} M_i$.

For any interaction subset $A = \{i_1, \dots, i_k\}$, the Khatri-Rao tier block $V_A^{(2)}$ under $H^{(2)}$ satisfies:

$$V_A^{(2)} = V_A^{(1)} M_A, \quad \text{where } M_A = M_{i_1} \otimes M_{i_2} \otimes \dots \otimes M_{i_k}. \quad (13)$$

Because M_A is invertible, $\text{range}(V_A^{(2)}) = \text{range}(V_A^{(1)})$. The empirical whitening step $\tilde{V}_A^{(2)} = W^{1/2} V_A^{(2)} = \tilde{V}_A^{(1)} M_A$ preserves column span. Applying the parent projection operator $P_\perp = I_r - Q_{\text{parents}} Q_{\text{parents}}^T$:

$$\tilde{V}_{A,\perp}^{(2)} = P_\perp \tilde{V}_A^{(2)} = P_\perp \tilde{V}_A^{(1)} M_A = \tilde{V}_{A,\perp}^{(1)} M_A. \quad (14)$$

Performing thin SVD yields $\text{range}(U_A^{(2)}) = \text{range}(\tilde{V}_{A,\perp}^{(2)}) = \text{range}(\tilde{V}_{A,\perp}^{(1)}) = \text{range}(U_A^{(1)})$. The orthogonal projection operator $P_{U_A} = U_A U_A^T$ onto this subspace is unique. Hence, $f_A^{(2)} = U_A^{(2)} (U_A^{(2)})^T y = U_A^{(1)} (U_A^{(1)})^T y = f_A^{(1)}$, completing the proof. $\square$

B ADDITIONAL ABLATION RESULTS

This appendix collects the detailed tables and figures supporting the five ablation studies summarized in Section 4. Tables 7–10 report the full numerical results, and Figures 4–5 provide their visual companions.

Table 7: Ablation 1: Single-variable contrast basis family comparison on CONNECT-4.

Contrast Family	cond(V_i)	cond(V_A)	gen	Main Cols (Wasted)	SVD Time	R^2 (Main)	R^2 (High)
Helmert (Default)	1.73e+00	3.17e+00		84 (2)	5.01s	0.449	0.717
One-Hot (Dummy)	1.00e+00	4.40e+00		126 (44)	7.96s	0.449	0.717
Centered Mean	6.20e+15	3.40e+15		126 (44)	7.72s	0.449	0.717
Fourier / WH	1.00e+00	3.56e+00		84 (2)	4.84s	0.449	0.717

Table 8: Ablation 2: Orthogonalization strategy comparison on CONNECT-4.

Strategy	Time (s)	Peak Mem (MB)	R^2	Ortho Viol (Max)	Ortho Viol (Mean)
Tier-Block Parent Projection (Default)	1.497	310.5	0.7295	1.44e-17	4.08e-22
Global Thin SVD	8.954	1346.7	0.7295	1.64e-02	7.35e-05
Unprojected Independent SVD	1.020	76.5	-534203.04	4.39e-01	8.66e-02
Column-wise Gram-Schmidt	1.385	870.7	0.7295	3.89e-18	2.72e-22

Table 9: Ablation 3: Empirical measure whitening comparison over 10 POKERHAND bootstraps.

Whitening Variant	R^2 ($L^2(P_n)$)	Weighted MSE	Unweighted MSE	Rank MAD	Importance CV
Baseline ($W^{1/2} = \text{diag}(\sqrt{P_n})$)	0.7810 $\pm$ 0.0040	0.0531 $\pm$ 0.0010	0.0576 $\pm$ 0.0008	0.920	0.0129
Identity ($W = I$)	0.7721 $\pm$ 0.0043	0.0553 $\pm$ 0.0010	0.0555 $\pm$ 0.0008	0.820	0.0105
Inverse Freq ($1/\sqrt{P_n}$)	0.7524 $\pm$ 0.0048	0.0601 $\pm$ 0.0012	0.0568 $\pm$ 0.0008	0.780	0.0112

Table 10: Ablation 5: Processing order metrics on CONNECT-4 ($r_{\text{low}} = 2000$).

Queue Ordering	R^2 @ Budget	Mean R^2 (Rank Eff.)	R^2 @ Rank 85	R^2 @ Rank 300	R^2 @ Rank 1000
Cardinality-First (Default)	0.664	0.548	0.450	0.481	0.553
Inverse-Cardinality	0.600	0.416	0.102	0.245	0.470
Variance-Prioritized	0.655	0.531	0.271	0.405	0.581
Random Order (Mean)	0.656	0.537	0.205	0.433	0.572

C IMPLEMENTATION DETAILS AND REPRODUCIBILITY

C.1 HYPERPARAMETERS AND HARDWARE SETUP

All experimental benchmarks were executed in Python 3.10 with PyTorch 2.2.0, CUDA 12.1, NumPy 1.26.4, and scikit-learn 1.4.0 on an NVIDIA RTX 4090 GPU (24GB VRAM) and Intel Xeon CPU host. Floating-point computations used `torch.float64` precision.

- **SVD Cutoff Threshold** (σ_{tol}): Default absolute singular value cutoff $\sigma_{\text{tol}} = 10^{-6}$, relative tolerance 10^{-9} .
- **Random Seeds**: Model training and bootstrap evaluation seeds fixed to 42 (resampling seeds 1000–1009).
- **Rank Budgets** (r_{low}): $r_{\text{high}} = 5000$ for POKERHAND and CONNECT-4; $r_{\text{high}} = 4000$ for DOTA2; $r_{\text{high}} = 10000$ for Binarized MNIST.

C.2 DATA PREPROCESSING AND BLACK-BOX MODELS

- **CAR EVALUATION & NURSERY**: Full hypergrids loaded directly; Target models: 100-tree Random Forest (`max_depth=None`) and 3-layer MLP (128 hidden units, ReLU, Adam optimizer).
- **MUSHROOMS**: 8,124 samples, 22 categorical features. Target model: XGBoost classifier (`n_estimators=100`, `max_depth=6`).
- **POKERHAND**: 25,000 training configurations. Target model: Tabular Transformer architecture with 4 multi-head attention layers and 128 embedding dimensions.

Sensitivity of Reconstruction Fidelity, Effective Rank, and Basis Conditioning to SVD Cutoff Threshold

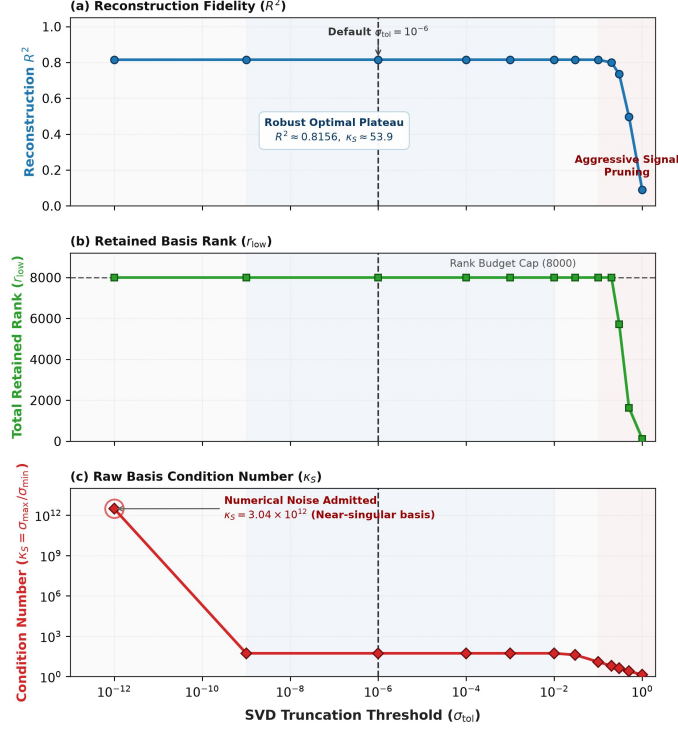


Figure 4: Sensitivity of TBOC-ANOVA metrics on the DOTA2 dataset across ten orders of magnitude of the SVD truncation threshold σ_{tol} . Panel (a) shows that reconstruction R^2 remains virtually constant across a broad plateau ($R^2 \approx 0.8156$) before dropping sharply under aggressive signal pruning for thresholds greater than 0.1. Panel (b) illustrates that total retained rank r_{low} reaches the 8000 budget cap across the robust optimal region and falls rapidly as genuine interaction directions are discarded. Panel (c) shows that an overly strict cutoff ($\sigma_{\text{tol}} = 10^{-12}$) admits near-singular numerical noise, inflating the condition number κ_S to 3.04×10^{12} , whereas thresholds within the optimal band (10^{-9} to 10^{-2}) yield well-conditioned bases ($\kappa_S \approx 53.9$). The default threshold of 10^{-6} achieves optimal reconstruction fidelity while suppressing numerical instability.

- **CONNECT-4:** 67,557 samples, 42 ternary categorical features. Target model: 100-tree Random Forest classifier predicting $P(\text{win})$.
- **DOTA2 RESULTS:** 102,944 game matches, 113 ternary features. Target model: 5-layer Deep MLP ($256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$).
- **Binarized MNIST:** 60,000 training images, thresholded at 0.5 into 784 binary pixels. Target model: 3-layer MLP classifier ($512 \rightarrow 256 \rightarrow 10$).

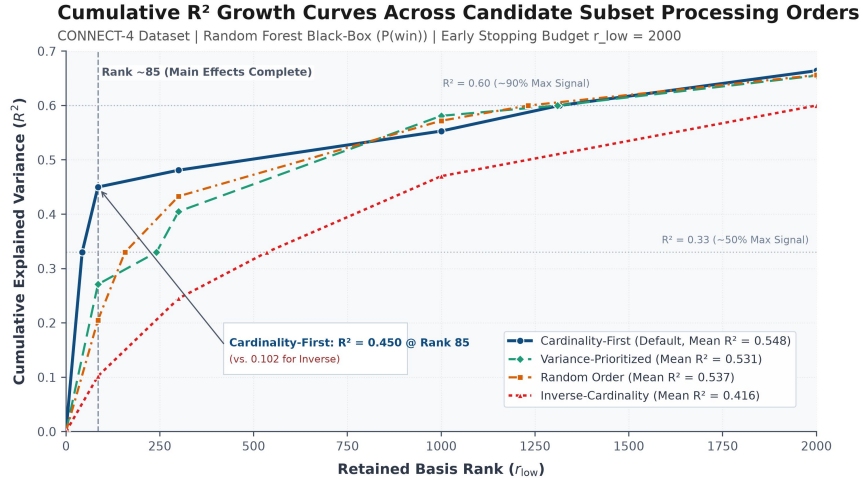


Figure 5: Cumulative explained variance growth as a function of retained basis rank across candidate subset processing strategies on the CONNECT-4 dataset under an early stopping budget of $r_{\text{low}} = 2000$. The default cardinality-first ordering rapidly captures the majority of black-box target variance at low ranks, achieving $R^2 = 0.450$ upon completing main effects at rank 85 compared to 0.102 for inverse-cardinality processing. This demonstrates the superior rank efficiency of prioritizing low-cardinality tiers in accordance with the sparsity-of-effects principle.